

The finiteness half of the mass gap is the nonvanishing of one smeared susceptibility

The requirement $m < \infty$ of the Jaffe–Witten statement reduces to an exact inequality between two ground-state quantities. For any gauge-invariant local scalar φ built from the spatially smeared field, with $\langle \varphi \rangle = 0$,

$$m \leq \frac{g^2 \hbar c}{2} \frac{\langle |D_\varphi|^2 \rangle_0}{\chi_\varphi}, \quad D_\varphi = \int d^3 y \frac{\delta \varphi(y)}{\delta A_i^a(0)}, \quad \chi_\varphi = \int d^3 z \langle \varphi(z) \varphi(0) \rangle_0^c,$$

in the continuum theory with a unique vacuum, whenever the finite-volume gaps and correlators converge. Both sides are intensive: the volume cancels between the numerator and the denominator, which is what makes the zero-momentum operator the right trial excitation and the plaquette of the upper-bound note the wrong one. The inequality is the energy-weighted sum rule, m bounded by the mean excitation energy carried by φ , so its content is: **if some local gauge-invariant observable has nonvanishing zero-momentum fluctuations in the vacuum, the theory has an excitation of finite energy.** Conversely $m = \infty$ forces every such susceptibility to vanish, that is, the vacuum is an eigenstate of every smeared gauge-invariant operator. The ultraviolet problem that defeated the plaquette is solved by smearing, and the natural smearing is the gradient flow, whose observables are renormalized; what is assumed, and stated as hypothesis (F1), is that the equal-time flowed correlators have continuum limits, which is not Lüscher’s four-dimensional theorem but its Hamiltonian counterpart. Dimensional analysis gives the perturbative behaviour $\langle |D|^2 \rangle / \chi \simeq c_\varphi / \sqrt{8t}$ at flow time t , so the bound reads $m \leq C g^2 \hbar c / \sqrt{8t}$ while perturbation theory applies; since the bound cannot fall below m , the flow time at which that scaling must break down is the correlation length $\hbar c / m$. In free Maxwell theory the scaling persists for all t and the bound correctly tends to $m = 0$. The inequality itself is the textbook single-mode (Feynman–Bijl) bound; what is specific here is the zero-momentum intensive form, the flow as the smearing that makes it cutoff-independent, and the identification with the finiteness half of the Millennium statement. Constants explicit; nothing promoted.

1. The continuum bound

Take the temporal-gauge Hamiltonian in the convention of G07 Proposition 7, with g dimensionless and $[A] = L^{-1}$:

$$H = \int d^3 x \left[\frac{g^2 c}{2\hbar} \Pi_i^a \Pi_i^a + \frac{\hbar c}{4g^2} F_{ij}^a F_{ij}^a \right], \quad [A_i^a(x), \Pi_j^b(y)] = i\hbar \delta^{ab} \delta_{ij} \delta^3(x - y),$$

obtained from $S = -\frac{\hbar}{4g^2} \int F_{\mu\nu}^a F^{\mu\nu} d^3 x c dt$ with $\Pi_i^a = (\hbar/g^2 c) \dot{A}_i^a$.

Proposition 1. Let O be a real functional of the spatial field A alone, gauge invariant, with $\langle O \rangle_0 = 0$ and $O\Omega \in D(H)$. Then

$$E_1 - E_0 \leq \frac{g^2 \hbar c}{2} \frac{\int d^3 x \left\langle \left| \frac{\delta O}{\delta A_i^a(x)} \right|^2 \right\rangle_0}{\langle O^2 \rangle_0}.$$

Proof. $O\Omega \perp \Omega$, so by min-max $E_1 - E_0 \leq \langle O\Omega, (H - E_0)O\Omega \rangle / \|O\Omega\|^2$, and $2\langle \Omega, O(H - E_0)O\Omega \rangle = \langle \Omega, [O, [H, O]]\Omega \rangle$ since $[O, [H, O]] = 2OHO - O^2H - HO^2$. The magnetic term commutes with O . For the electric term, $[\Pi_i^a(x), O] = -i\hbar \delta O / \delta A_i^a(x)$ and

$$[O, [\Pi_i^a \Pi_i^a, O]] = -2[\Pi_i^a, O]^2 = 2\hbar^2 \left| \frac{\delta O}{\delta A_i^a} \right|^2,$$

so $[O, [H, O]] = \frac{g^2 c}{2\hbar} \cdot 2\hbar^2 \int d^3x |\delta O / \delta A|^2 = g^2 \hbar c \int d^3x |\delta O / \delta A|^2$. Divide by $2\langle O^2 \rangle$. $\square$

Only the electric term contributes: the bound measures how much the trial observable is disturbed by the conjugate field, which is the same quantity that Section 2 of the action-floor note identifies as the carrier of $\hbar$.

2. Zero momentum makes the bound intensive

Let $\varphi(x)$ be a gauge-invariant local scalar with $\langle \varphi \rangle_0 = 0$ and take $O = \int_V d^3y \varphi(y)$ in a periodic box of volume V . Write

$$D_i^a(x) = \frac{\delta O}{\delta A_i^a(x)} = \int_V d^3y \frac{\delta \varphi(y)}{\delta A_i^a(x)}.$$

By translation invariance of the ground state, $\langle |D(x)|^2 \rangle_0$ does not depend on x , so the numerator of Proposition 1 is $V \langle |D|^2 \rangle_0$, while $\langle O^2 \rangle_0 = \int_V \int_V \langle \varphi(y) \varphi(y') \rangle_0^c = V \chi_\varphi$ with $\chi_\varphi = \int d^3z \langle \varphi(z) \varphi(0) \rangle_0^c$. The volume cancels:

Corollary 2. For every such φ and every V ,

$$\Delta_V \leq \frac{g^2 \hbar c}{2} \frac{\langle |D_\varphi|^2 \rangle_0}{\chi_\varphi},$$

and if $\Delta_V \rightarrow m$ and both ground-state quantities converge, the same bound holds for m .

Dimensions: $\delta / \delta A(x)$ carries $L^{-3} / [A] = L^{-2}$, so $\langle |D|^2 \rangle = [\varphi]^2 L^2$ and $[\chi] = [\varphi]^2 L^3$; the ratio is an inverse length and $g^2 \hbar c$ times it is an energy. The plaquette of the upper-bound note fails precisely because it is a single cell rather than a zero-momentum sum: its variance is an ultraviolet quantity that vanishes in the continuum limit while its gradient does not.

3. Spectral reading and two checks

Inserting a complete set of eigenstates,

$$\langle \Omega, [O, [H, O]] \Omega \rangle = 2 \sum_n (E_n - E_0) |\langle n | O | \Omega \rangle|^2, \quad \langle O^2 \rangle_0 = \sum_n |\langle n | O | \Omega \rangle|^2,$$

so Proposition 1 states that m is at most the mean excitation energy weighted by the spectral weight of O : the f -sum rule. Two consequences.

Conserved observables. If $[H, O] = 0$ the numerator vanishes and the bound reads $\Delta \leq 0$. With a unique ground state this is consistent only if $\langle O^2 \rangle_0 = 0$: a conserved gauge-invariant observable cannot fluctuate in a nondegenerate vacuum, since $O\Omega$ would be a second state of energy E_0 . The bound therefore detects vacuum degeneracy, and is vacuous exactly in that case.

Free Maxwell. Here $m = 0$ and the bound is a positive number for every flow time; Section 5 shows it tends to zero as the smearing grows, which is the correct behaviour for a gapless theory and a check that the bound is not empty.

4. What must be smeared, and the hypothesis

A local operator at a point has divergent gradient in the continuum, so φ must be built from a smeared field. The gauge-covariant smearing is the gradient flow: on the spatial field at fixed time,

$$\partial_s B_i = D_j G_{ji}, \quad B_i|_{s=0} = A_i, \quad G_{ij} = \partial_i B_j - \partial_j B_i + [B_i, B_j],$$

the three-dimensional, equal-time counterpart of Lüscher's flow (arXiv:1006.4518v3, equations (1.1)–(1.2), passage level via the local companion), with smearing radius $\sqrt{8t}$ at flow time t and the lattice form given by his equation (1.4). Take $\varphi = \varphi_t = \frac{1}{4} G_{ij}^a G_{ij}^a|_{s=t} - \langle \cdot \rangle_0$.

Lüscher's renormalization statement is for the four-dimensional flow in the Euclidean functional integral: expectation values of local gauge-invariant expressions in the flowed field are finite without further renormalization. The object needed here is the equal-time Hamiltonian correlator of the spatially flowed field, which his theorem does not cover. State it as a hypothesis:

(F1) For fixed flow time $t > 0$, the equal-time ground-state quantities χ_{φ_t} and $\langle |D_{\varphi_t}|^2 \rangle_0$ of the lattice theory converge, as $a \rightarrow 0$ along the scaling curve, to finite limits, with $\chi_{\varphi_t} > 0$.

On any fixed lattice both quantities are finite and positive and Corollary 2 holds unconditionally, so (F1) is entirely a statement about the continuum limit. Its first half is the flow's renormalization property in the Hamiltonian framework; its second half, $\chi > 0$, is nontriviality: the vacuum is not an eigenstate of the flowed energy density.

Proposition 3 (the finiteness half). Assume the continuum theory exists with a unique vacuum and translation invariance, that the finite-volume gaps converge to m , and (F1) for one flow time t . Then

$$m \leq \frac{g^2 \hbar c}{2} \frac{\langle |D_{\varphi_t}|^2 \rangle_0}{\chi_{\varphi_t}} < \infty.$$

This is the $m < \infty$ clause of the Jaffe–Witten statement, reduced to the positivity of one susceptibility. The remaining clause, $m > 0$, is untouched: nothing above excludes $m = 0$.

5. The flow time, and where the perturbative scaling must fail

At flow time t the only length in the smeared operator is $\sqrt{8t}$. In a regime where the flowed correlators are governed by that length alone, $[\varphi_t] = L^{-4}$ gives $\chi \simeq k_1 t^{-5/2}$ and $\langle |D|^2 \rangle \simeq k_2 t^{-3}$, hence

$$\frac{\langle |D_{\varphi_t}|^2 \rangle}{\chi_{\varphi_t}} \simeq \frac{k_2}{k_1} \frac{1}{\sqrt{8t}}, \quad m \leq C g^2 \frac{\hbar c}{\sqrt{8t}}.$$

The pure numbers k_1, k_2 are not computed here. Two readings.

Free theory. In free Maxwell theory the scaling holds for every t because there is no other length, and the bound tends to $0 = m$: correct, and the flow time is the only resolution scale.

Interacting theory. A positive m cannot be beaten by any t , so the perturbative scaling must fail once $C g^2(t) \hbar c / \sqrt{8t}$ reaches m , that is at

$$\sqrt{8t_*} \simeq C g^2 \frac{\hbar c}{m},$$

the correlation length up to the coupling. The bound therefore contains its own consistency condition: the flow smooths the field only down to the confinement scale, beyond which the flowed correlators are no longer controlled by $\sqrt{8t}$. This is the Hamiltonian counterpart of the crossover recorded in the valley note, where the small-volume description fails when the zero-mode energy reaches $\hbar c/L$.

Approach to m . As t grows the smeared operator couples preferentially to states of spatial extent $\gtrsim \sqrt{8t}$, so the spectral weights of Section 3 concentrate on the lowest states and the bound is expected to decrease toward m . Monotonicity is not proved here; it would make the family of bounds a convergent sequence rather than a single estimate.

6. Status

claim	status
Proposition 1, the bound with explicit constants	proved
Corollary 2, intensive zero-momentum form	proved
f -sum rule reading, conserved-observable check	proved
finiteness of the lattice bound at fixed a	proved

claim	status
(F1), continuum limit of the flowed equal-time correlators	hypothesis; the flow's renormalization in the Hamiltonian framework
$\chi > 0$	hypothesis; nontriviality
$\langle D ^2 \rangle / \chi \simeq c / \sqrt{8t}$	dimensional, coefficients not computed
$m > 0$	untouched

The single-mode bound and the use of smeared operators as glueball interpolators are standard; the statement being recorded is the reduction of the Millennium finiteness clause to the positivity of one flowed susceptibility, with the constants and hypotheses written out.

7. Consequence for STATE

The upper side of the problem is now in the same shape as the lower side: an exact inequality at fixed cutoff, plus one renormalization hypothesis for the continuum. The two hypotheses are different in kind. (F1) concerns flowed observables, which are expected to be renormalized by Lüscher's four-dimensional result and whose Hamiltonian counterpart is a bounded, self-contained question; the dressed-vacuum problem of the Schur note concerns the state itself. The next steps this suggests, in order: compute k_1, k_2 in free Maxwell theory to make Section 5 quantitative and check the free limit; then state the Hamiltonian flow-renormalization question as its own target, since both the upper-side (F1) and the lower-side dressing are instances of it.